CSE4203: Computer Graphics

Lecture – 8 (Part - A)

Fractal Geometry

Outline

- Fractal
- Fractal Dimension
- The Cantor Set
- Koch Snowflake
- Sierpinski Triangle
- Mandelbrot Set

Fractal

- A **Fractal** is a geometric shape generated using set of recursive rules
- Fractal share a **self-similarity** property where part of fractal resembles the whole fractal
- The property is often seen in nature, e.g clouds, plants and landscapes etc.
- We will look at methods used to generate some popular fractals in this lecture

Credit:https://www.youtube.com/@mathematicsofcomputergraph6990

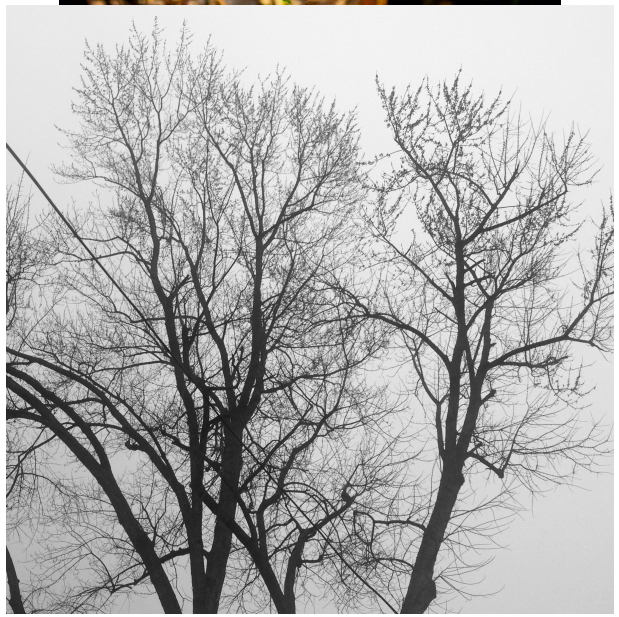
Example of Fractals







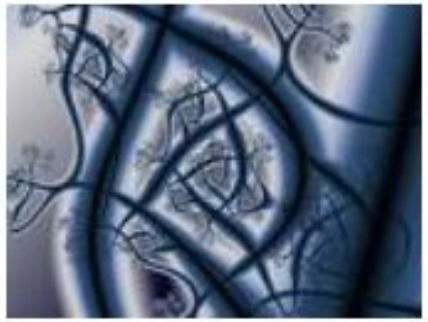


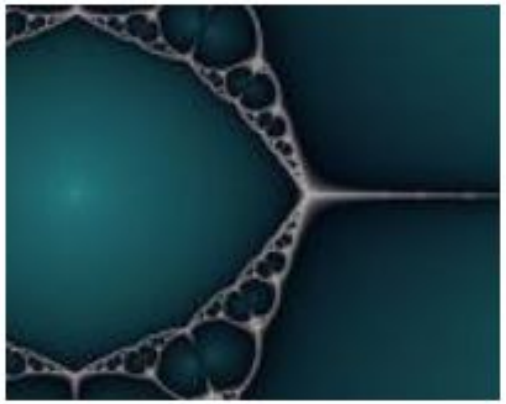


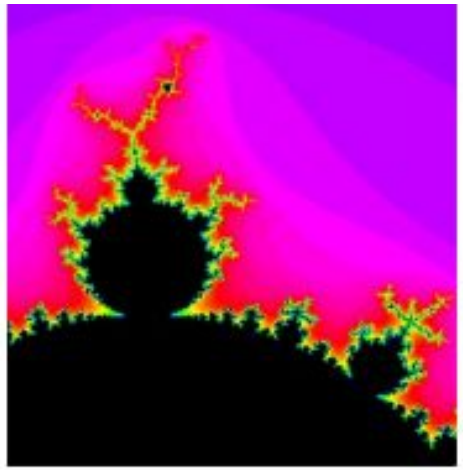


Credit:Internet

Artificial Fractals













Credit:Internet

The Cantor Set (1/6)

- A simplest example of a fractal is a Cantor Set devised by George Cantor
- Starting with a straight line at n = 0, we remove middle third of each line at each iteration
- As n gets larger, line segments gets smaller

The Cantor Set (2/6)

- A simplest example of a fractal is a Cantor Set devised by
- n = 0 

The Cantor Set (3/6)

- A simplest example of a fractal is a Cantor Set devised by
- n = 0
- n = 1



The Cantor Set (4/6)

- A simplest example of a fractal is a Cantor Set devised by

- n = 0
- n = 1
- n = 2



The Cantor Set (5/6)

- A simplest example of a fractal is a Cantor Set devised by

- n = 0
- n = 1
- n = 2
- n = 3



The Cantor Set (6/6)

- A simplest example of a fractal is a Cantor Set devised by



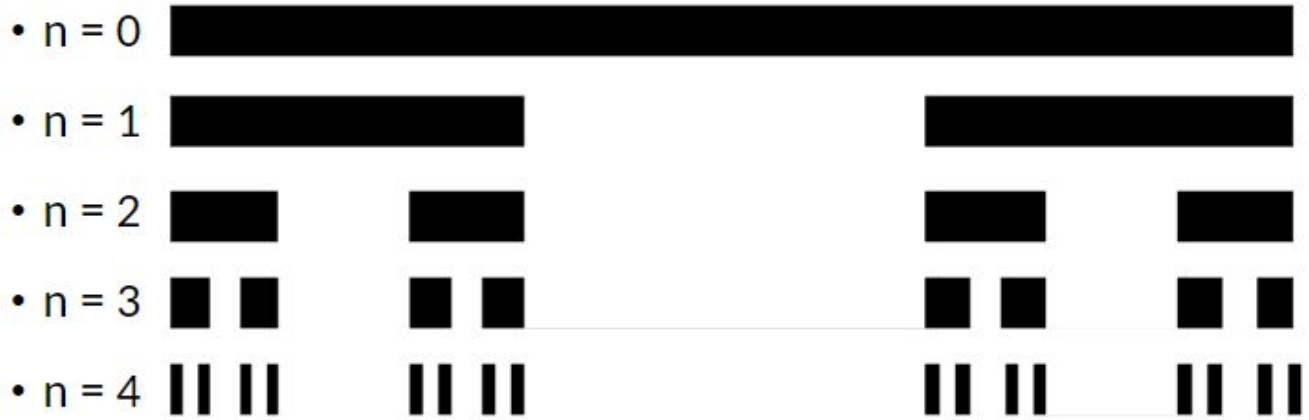
Fractal Dimension

- The fractal dimension is a measure of the complexity of a fractal
- Let N_n be the number of new elements added at iteration n and ϵ be the scaling factor of the size of new elements then the fractal dimension is

$$D = -\lim_{n\to\infty} \frac{\log(N_n)}{\log(\epsilon_n)}$$

- For a 2D fractal $D \in [1, 2]$ and for a 3D fractal $D \in [2, 3]$

Dimension of Cantor Set (1/2)



Line segments is related to iterations

iterations	no. line segments	length of segment
0	1	1
1	2	$\frac{1}{3}$
2	4	$\frac{1}{9}$
3	8	$\frac{1}{27}$
⋮	⋮	⋮
n	2^n	3^{-n}

Dimension of Cantor Set (2/2)

Using

$$D = -\lim_{n\to\infty} \frac{\log(N_n)}{\log(\epsilon_n)}$$

then the fractal dimension of Cantor set is

$$D = -\lim_{n\to\infty} \frac{\log(2^n)}{\log\frac{1}{3^n}} = -\lim_{n\to\infty} \frac{n\log 2}{n\log\frac{1}{3}} = -\lim_{n\to\infty} \frac{\log 2}{\log 3^{-1}}$$

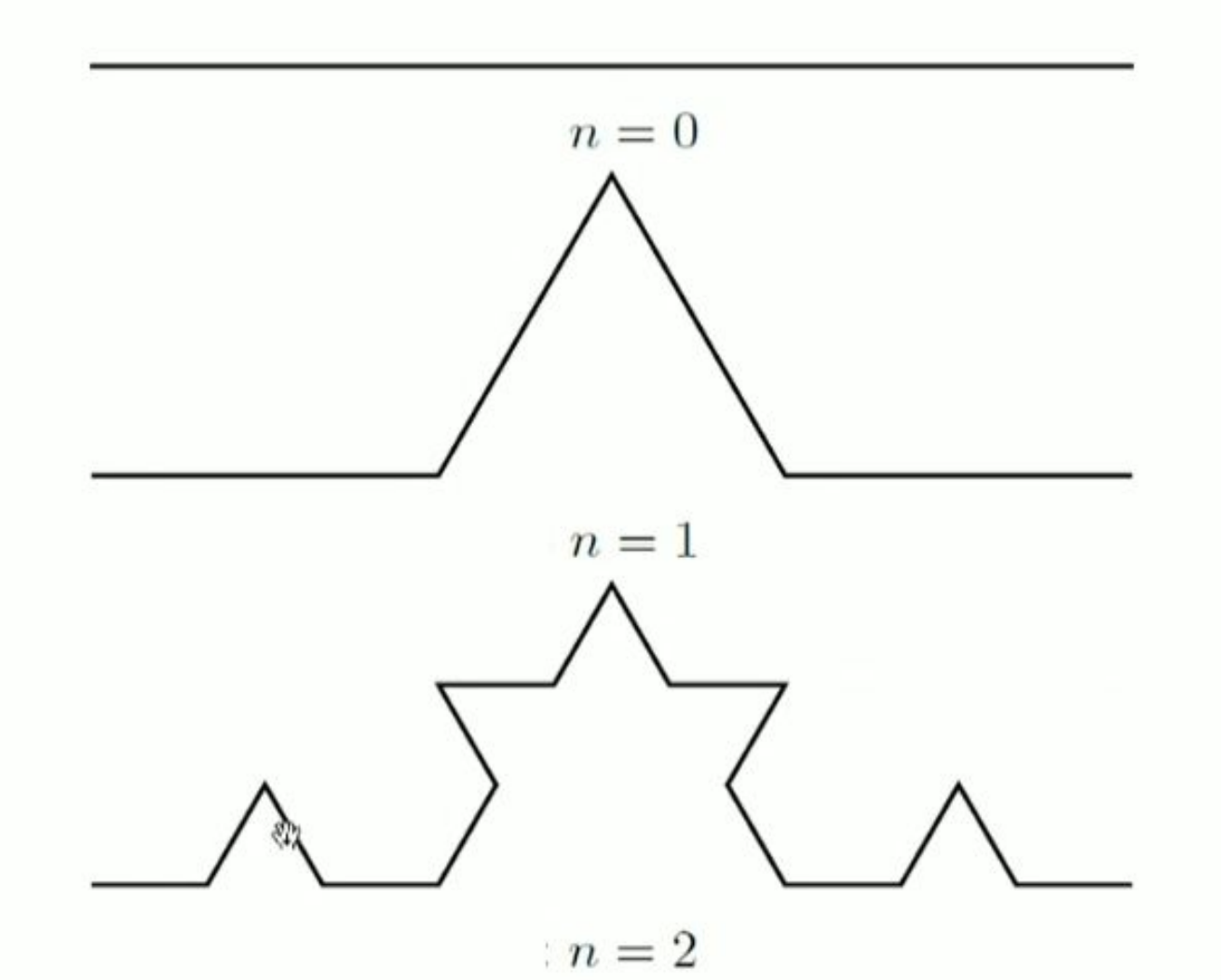
$$= \lim_{n\to\infty} \frac{\log 2}{\log 3}$$

$$= 0.6309\ldots$$

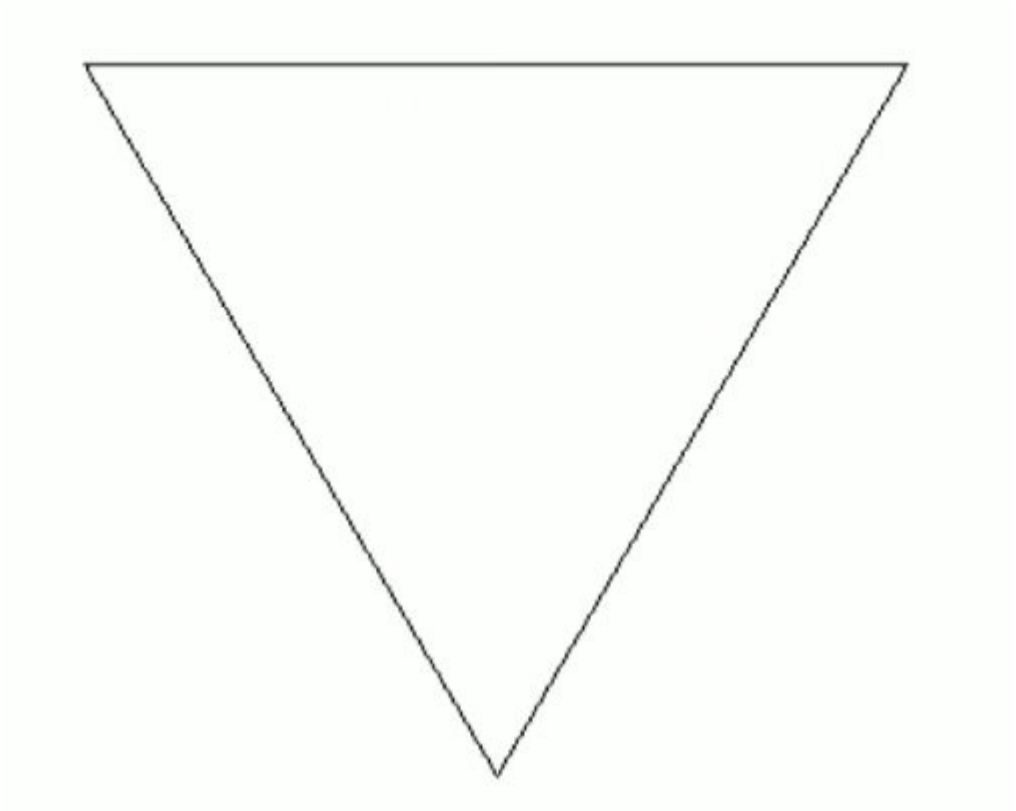
The Koch Snowflake (1/7)

- The Koch Snowflake named after Helge von Koch is a shape with a finite area bounded by infinite circumference
- The rules for generating a Koch Snowflake start with a equilateral triangle at stage = 0
  - Divide each line into three lines of equal length
  - Place an equilateral triangle so that the base fits along the middle length
  - Remove the middle length that is base of the triangle

The Koch Snowflake (2/7)

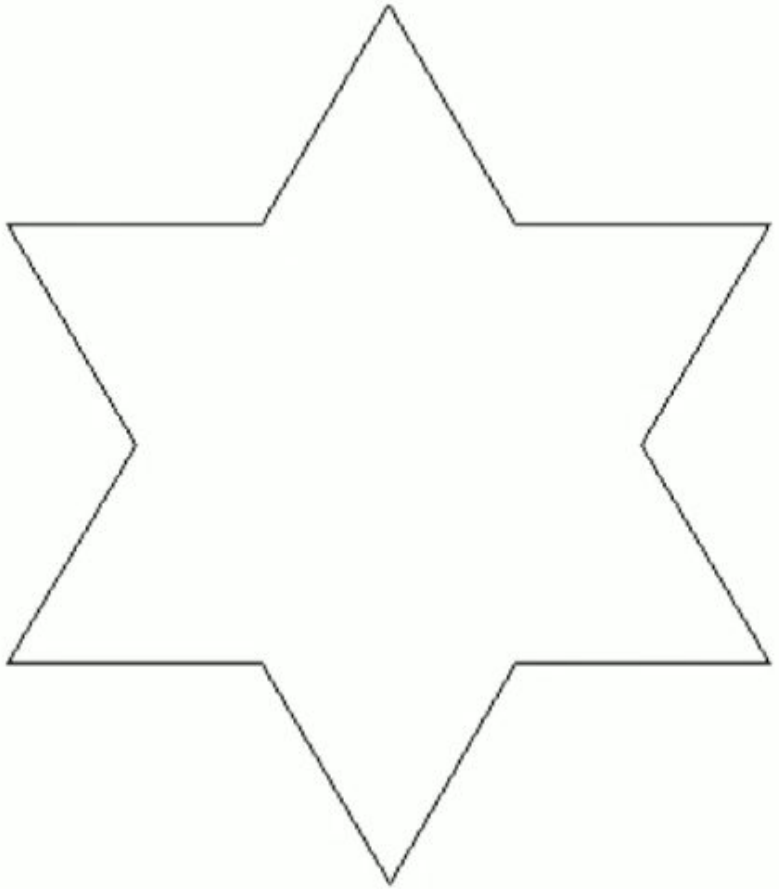


The Koch Snowflake (3/7)



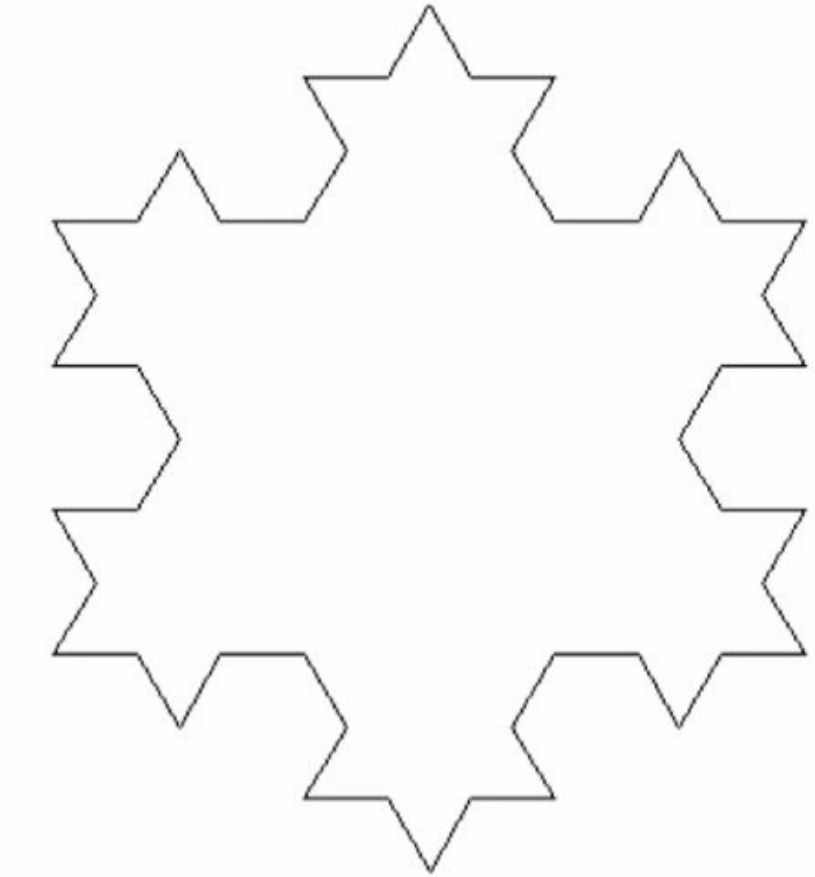
stage	no of sides	length of sides
0	3	1

The Koch Snowflake (4/7)



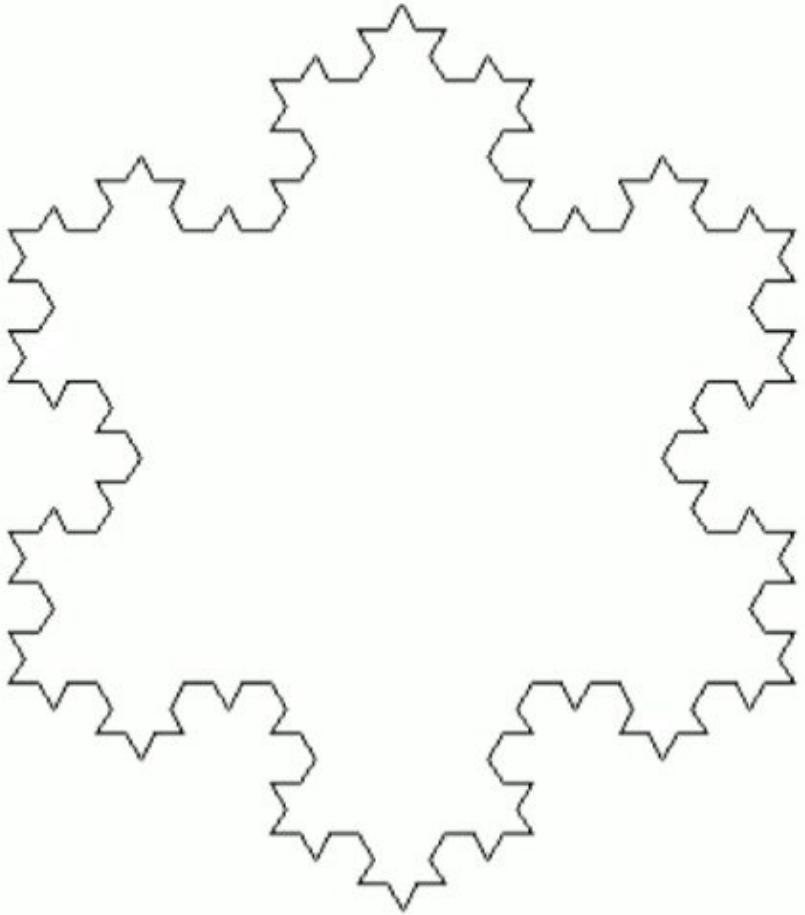
stage	no of sides	length of sides
1	3 x 4	1/3

The Koch Snowflake (5/7)



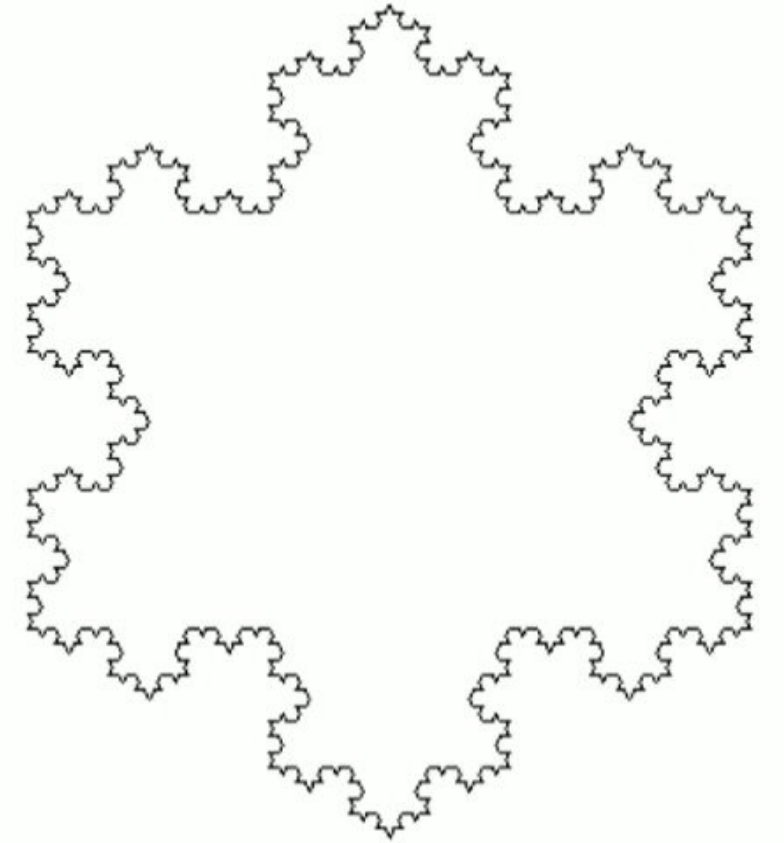
stage	no of sides	length of sides
2	3 x 4 x 4	$1/3^2$

The Koch Snowflake (6/7)



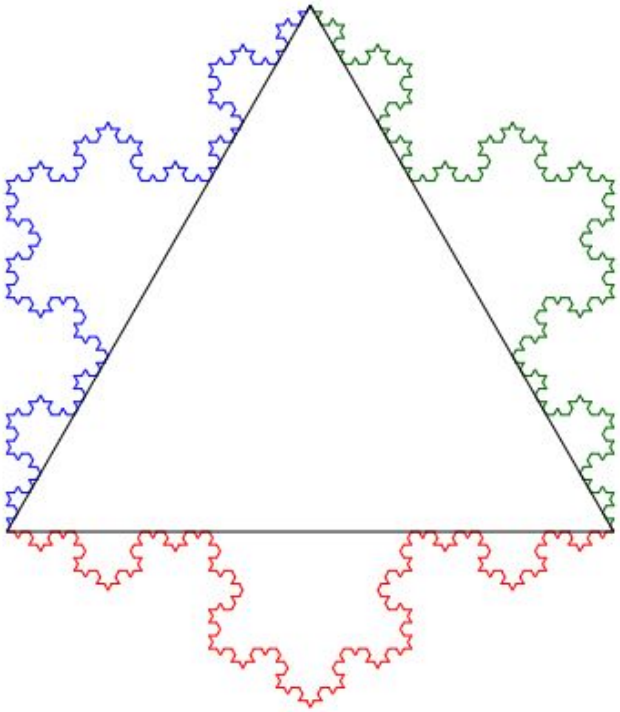
stage	no of sides	length of sides
3	3 x 4 x 4 x 4	$1/3^3$

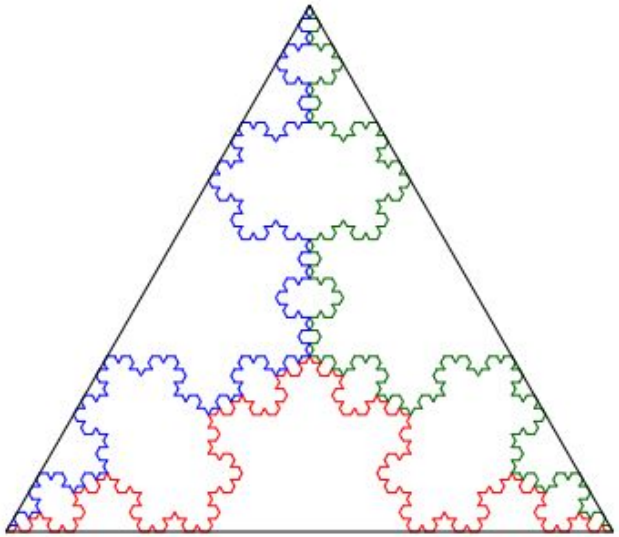
The Koch Snowflake (7/7)

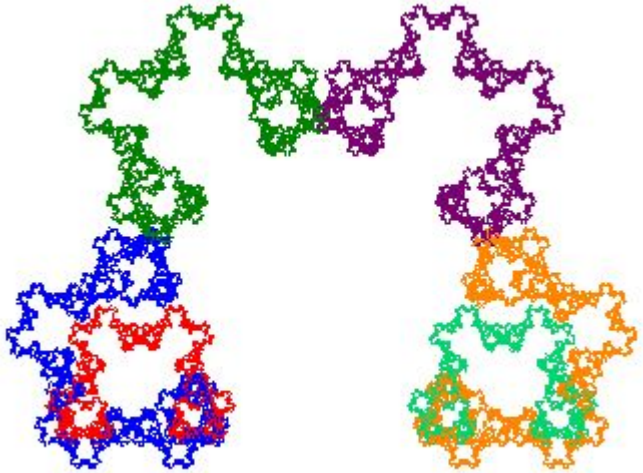


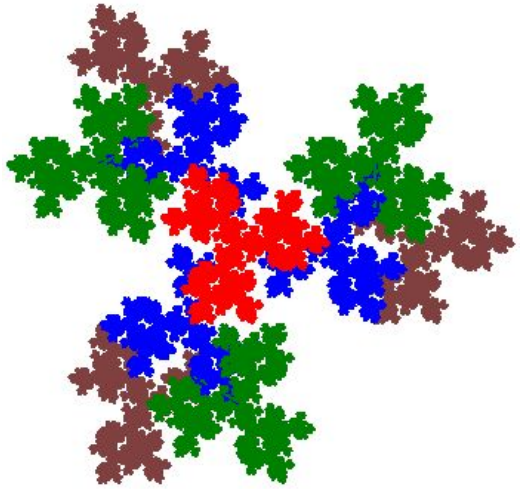
stage	no of sides	length of sides
4	3 x 4 x 4 x 4 x 4	$1/3^4$

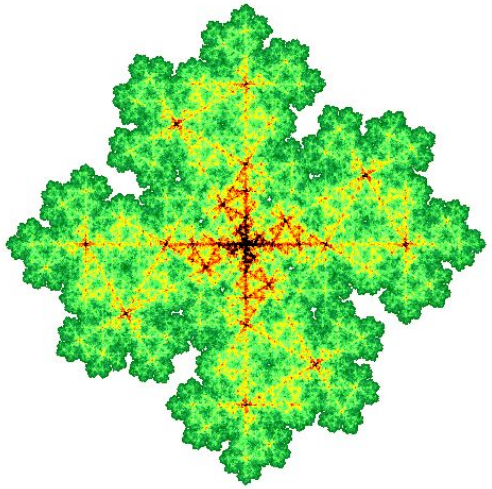
Colorful Koch Snowflakes

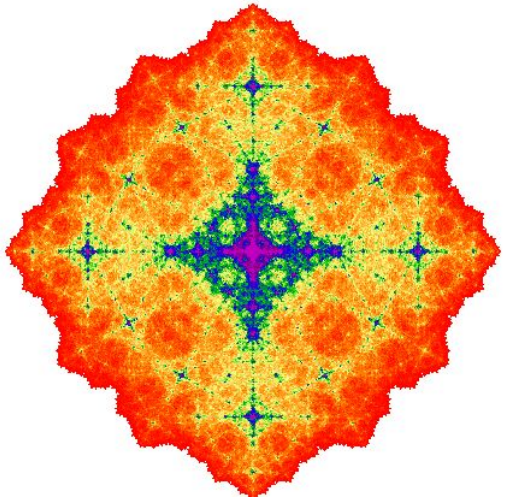






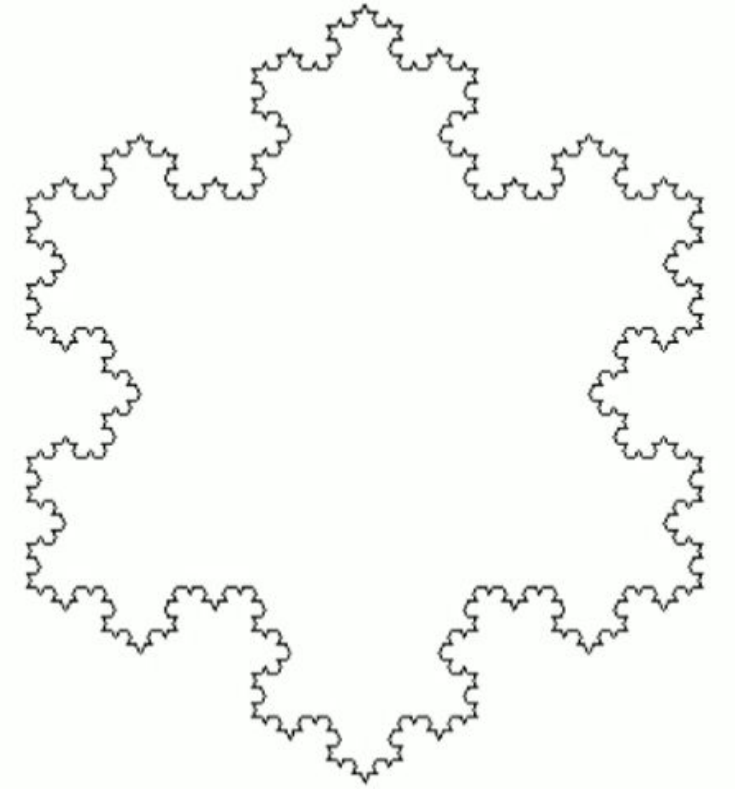






[Koch Curve (agnesscott.org)]

Fractal Dimension of Koch Snowflake (1/2)



stage	no of sides	length of sides
0	3	1
1	3 x 4	1/3
2	3 x 4 x 4	$1/3^2$
3	3 x 4 x 4 x 4	$1/3^3$
4	3 x 4 x 4 x 4 x 4	$1/3^4$
…	…	…
n	3×4^n	$1/3^n$

Fractal Dimension of Koch Snowflake (2/2)

Using

$$D = -\lim_{n\to\infty} \frac{\log(N_n)}{\log(\epsilon_n)}$$

then the fractal dimension of Koch Snowflake is

$$D = -\lim_{n\to\infty} \frac{\log\,[3(4^n)]}{\log\left[\frac{1}{3^n}\right]} = -\lim_{n\to\infty} \frac{\log(3) + n\log 4}{-n\log(3)}$$

$$= \lim_{n\to\infty} \frac{\log(3)}{n\log(3)} + \frac{n\log 4}{n\log(3)} = \frac{\log(4)}{\log(3)}$$

$$= 1.2619\ldots$$

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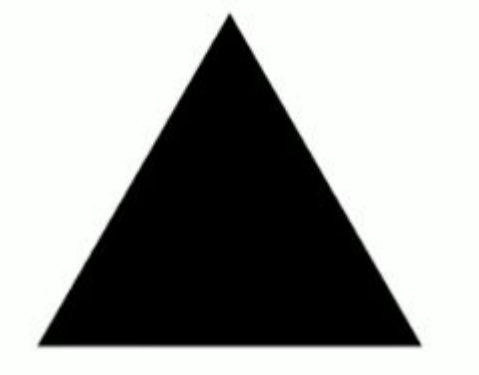
$$= \lim_{n\to\infty} \frac{\log(3)}{n\log(3)} + \frac{n\log 4}{n\log(3)} = \frac{\log(4)}{\log(3)}$$

$$= 1.2619\ldots$$

Sierpinski Triangle (1/5)

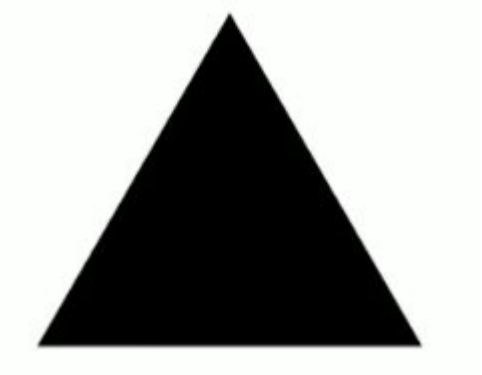
- Attributed to Waclaw Sierpinski, the Sierpinski has also been seen in the medieval mosaics
- It starts with a filled equilateral triangle at stage 0 and the rules are:
  - For each filled triangle, make an unfilled copy
  - Scale the copy by half and invert it
  - Place the copy in the center of the filled triangle

Sierpinski Triangle (2/5)



stage 0

Sierpinski Triangle (3/5)

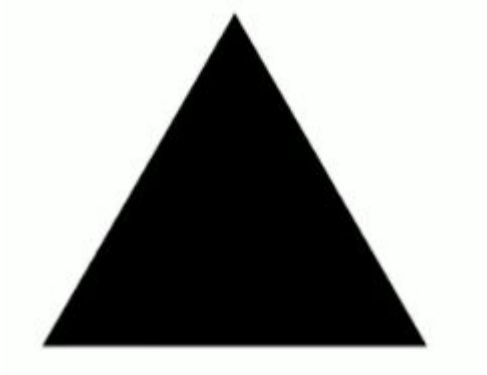


stage 0

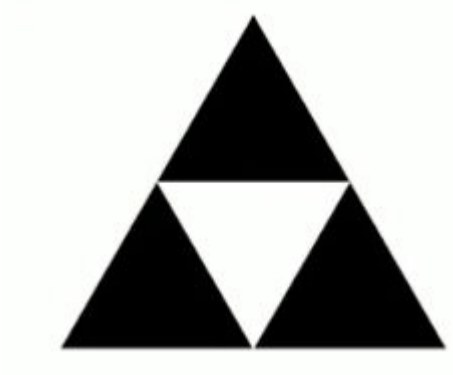


stage 1

Sierpinski Triangle (4/5)



stage 0

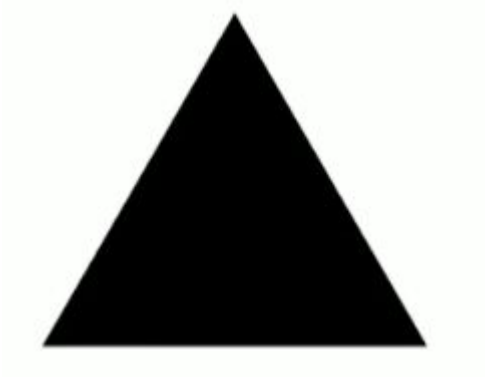


stage 1

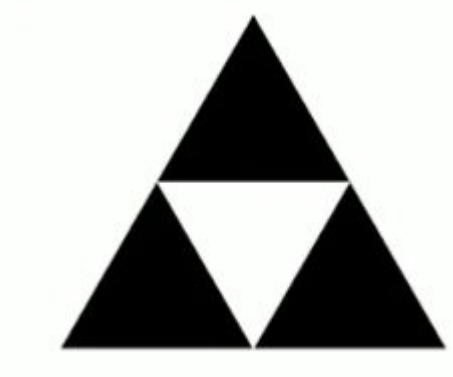


stage 2

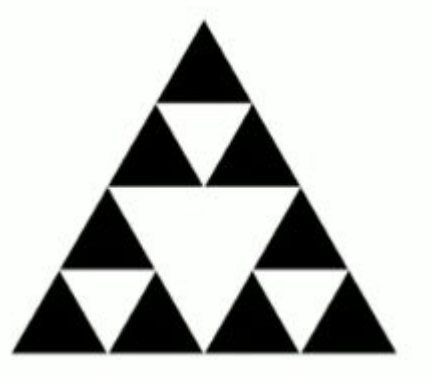
Sierpinski Triangle (5/5)



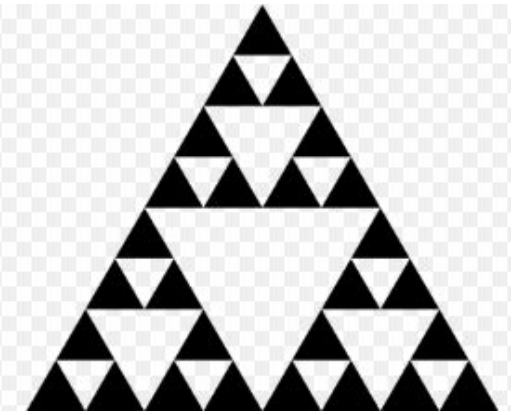
stage 0



stage 1



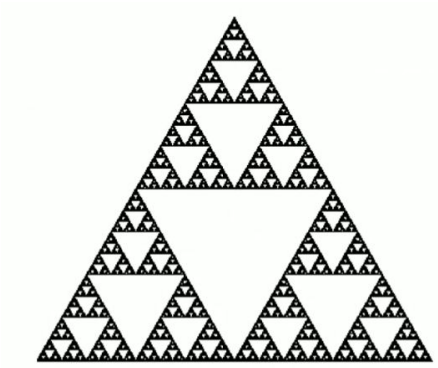
stage 2



stage 3

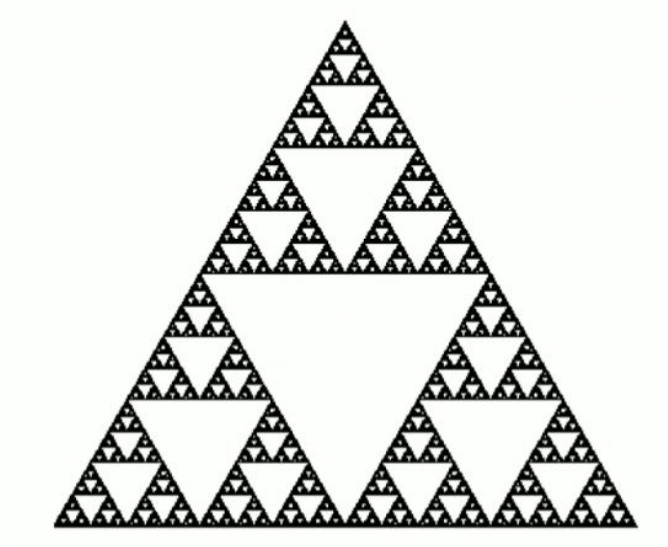


stage 5



stage 6

Fractal Dimension of Sierpinski Triangle



$$D = \frac{\log(3)}{\log(2)}$$

$$= 1.585 \ldots$$

How?

[Zoom into Sierpoinski Triangle]

Further Study

- https://www.youtube.com/watch?v=bEXW7V9mATU
- https://www.youtube.com/watch?v=NGMRB4O922I&ab_channel=Numberphile